

Verification and Approximation Study for *Generalized Cluster Dynamics* — Campaign Results

RadCluster_2.1

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Abstract

This report collects the results of the verification campaign run in answer to the reviewer’s §6.2 / Figure 8 critique. It reports the six experimentally observable parameters for every campaign configuration, against the EUROFER97 measurements where a comparison is meaningful and against the fully discrete solution for the closure verification. Two defects found *by* the campaign — a physics channel missing from one solver path, and a dose ladder that scored the wrong dose — are documented alongside the numbers they invalidated, because they bear directly on what the tables may be used to claim.

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1 Campaign Parameters

All runs share the EUROFER97 host declaration, fission cascade spectrum, $G = 1 \times 10^{-7}$ dpa/s, 330 °C, `full_system` solver mode, GMRES with the Woodbury preconditioner, `atol` = 10^{-20} , and $i_{\text{mobile}} = v_{\text{mobile}} = 5$. Table 1 lists what differs.

The EUROFER97 neutron measurements at 300 °C to 350 °C (Table 2) are the experimental column throughout. Their dose coverage matters: the loop data begin at 14.6 dpa, which is why Tables 1–3 are scored at 15.72 dpa and why Table 4, which runs to 2 dpa, cannot be compared with them at all.

Table 1: Campaign configuration. N_{eq} is the main block; the $\langle 100 \rangle$ loop block is carried separately and adds a further I equations in discrete mode (so run D integrates 12 006, not 8 006).

Block	Equations	I	V	i_{discrete}	I_{bin}	Dose (dpa)	Scored at
Table 1 (B1–B4)	bin_moment	80 000	20 000	6400–100	7–18	40	15.72 dpa
Table 2	bin_moment	80 000	20 000	100	10–40	40	15.72 dpa
Table 3	bin_moment	80 000	20 000	100	18	40	15.72 dpa
Table 4 (D)	discrete	4 000	4 000	—	—	2	2.000 dpa
Table 4 (C0–C4)	bin_moment	4 000	4 000	1600–5	2–19	2	2.000 dpa

Table 2: EUROFER97, neutron, 300 °C to 350 °C. Measured ranges over all sources in docs/Database.

Population	n	Density (10^{21} m^{-3})	Diameter (nm)
$\frac{1}{2}\langle 111 \rangle$ loops	13	0.30 – 34.0	3.3 – 35.0
$\langle 100 \rangle$ loops	10	2.24 – 12.0	2.8 – 48.0
Cavities	6–8	0.32 – 6.30	1.6 – 16.0
Dose coverage: loops 14.6 dpa to 32 dpa, cavities 2.7 dpa to 32 dpa			

Table 3: Closure convergence, $I = 80\,000$, $V = 20\,000$, scored at 15.72 dpa. B4 is the tracked reference run. B1 is reported as *did not reach* under the rule that a run missing its comparison dose never appears as a metric.

Rung	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
B1 (i_d 6400)	~ 8030	<i>did not reach</i> — 0.079 dpa of 40 after 18.9 h						
B2 (i_d 1600)	2048	3.449	4.892	4.156	6.001	2.233	2.232	0.069
B3 (i_d 400)	562	3.512	4.509	4.090	5.868	2.237	2.234	0.068
B4 = production	189	3.498	4.637	4.715	5.939	3.609	2.890	0.060
<i>Experiment</i>		≥ 0.30	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16	—

1.1 Table 1 — Closure convergence at the production domain

1.2 Table 2 — Intra-bin closure and tolerance

Table 4: One knob changed from B4 at a time, scored at 15.72 dpa.

Column	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
B4 (reference)	189	3.498	4.637	4.715	5.939	3.609	2.890	0.060
$P = 1$ constant	150	2.566	15.580	5.335	45.676	11.361	2.751	0.158
$P = 3$ lognormal	228	<i>did not reach</i> — 5.4×10^{-4} dpa of 40; 2.9 h on one output step						
$I_{\text{bin}} 10$	171	3.543	4.282	4.021	5.861	2.900	2.340	0.070
$I_{\text{bin}} 40$	233	<i>did not reach</i> — 5.70 dpa of 40						
rtol 1×10^{-4}	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
rtol 1×10^{-6}	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
<i>Experiment</i>	≥ 0.30		3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16	—

1.3 Table 3 — Helium

Table 5: Helium treatment, scored at 15.72 dpa. Model-to-model: the fusion arms have no matching EUROFER97 dataset.

Column	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)	δ_{FP}
fission QSS	189	3.533	4.354	4.153	5.919	3.253	2.434	0.068
fission dynamic	190	3.533	4.354	4.153	5.919	3.253	2.434	0.068
fusion QSS	208	2.693	14.030	4.155	51.921	1.360	0.699	0.158
fusion dynamic	—	<i>did not reach</i> — 0.052 dpa of 40 after 13.9 h						
fusion Case 1/2	—	blocked on the §3.1 <code>he_model</code> unwelding (not implemented)						

1.4 Table 4 — Closure verification against the exact solution

Deviations here are quoted against run D, the fully discrete solution, not against experiment. Both arms run with `LOOP_COAL = 0` so that they solve identical equations; see §4.

Table 6: Closure verification, $I = V = 4000$, `LOOP_COAL = 0` in both arms, scored at 2.0000 dpa. Percentages are deviations from run D. N_{eq} is the main block, which is what the solver reports; *total* adds the appended $\langle 100 \rangle$ loop block that CVODE also integrates (I equations in discrete mode, $i_{\text{discrete}} + 2I_{\text{bin}}$ in bin-moment mode). The figures in §3 are labelled with the total.

Rung	N_{eq}	total	N_{111} (10^{21})	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)
D (exact)	8006	12006	12.568	1.065	5.741	2.948	2.544
C0 (finer than C1)	1226–3214	—	did not reach — the no-coarsening variant is markedly stiffer the more of the frozen spectrum is resolved discretely; C1 ran in 12 s, $i_d = 600$ –1600 did not complete				
C1 (1/10)	830	1242	12.335 (−1.9%)	1.044 (−2.0%)	5.690 (−0.9%)	2.555 (−13.3%)	2.273 (−10.6%)
C2 (1/40)	244	364	12.144 (−3.4%)	1.044 (−2.0%)	5.681 (−1.0%)	2.574 (−12.7%)	2.279 (−10.4%)
C3 (1/160)	108	161	7.027 (−44.1%)	1.018 (−4.4%)	5.349 (−6.8%)	3.219 (+9.2%)	2.451 (−3.6%)
C4 (1/800)	86	129	6.142 (−51.1%)	0.953 (−10.5%)	5.678 (−1.1%)	6.575 (+123.0%)	2.660 (+4.6%)
C4, $P = 1$	51	94	17.498 (+39.2%)	0.059 (−94.5%)	10.755 (+87.3%)	20.929 (+610%)	2.632 (+3.5%)

1.5 Table 4b — The extended rungs against experiment

C2–C4 were rerun to the production horizon so that their curves reach the dose where the EUROFER97 measurements actually sit. At 15.72 dpa they can therefore be compared with experiment directly — something Table 4 cannot do, since it is scored at 2 dpa where there are no loop data at all. Run D has no counterpart here: the discrete arm could not be carried past 0.014 dpa, so **beyond 2 dpa these rungs have no exact reference**. They show dose dependence and the approach to the measured band, not verified accuracy.

Table 7: The no-coarsening rungs carried to 40 dpa, scored at 15.72 dpa against the EUROFER97 measured ranges. *Not* a verification: there is no exact solution at this dose.

Rung	N_{eq}	N_{111} (10^{21})	d_{111} (nm)	N_{100} (10^{21})	d_{100} (nm)	N_{cav} (10^{20})	d_{cav} (nm)
C2 (1/40)	244	13.092	1.722	6.916	5.045	1.951	2.135
C3 (1/160)	108	6.770	1.487	6.392	4.770	2.235	2.230
C4 (1/800)	86	5.276	1.476	5.950	4.748	3.431	2.484
<i>Experiment</i>	—	0.30–34.0	3.3–35	2.24–12.0	2.8–48	3.2–63	1.6–16

All three land inside the measured $\langle 100 \rangle$ density and diameter ranges and inside the cavity diameter range. N_{111} sits above the measured band and d_{111} below it — the signature of the disabled coarsening channel, which is what pins the $\frac{1}{2}\langle 111 \rangle$ mean size at the mobility cutoff and inflates its density. That is the expected consequence of running the no-coarsening variant, not an independent finding.

1.6 Table 5 — Null knobs

Table 5 (preconditioner and thread-count null tests) was not run. Its columns measure wall-clock as the dependent variable, so they require an otherwise idle machine, and the campaign machine was occupied throughout. The evidence already in the repository stands in its place: from `compare.linsol.results.json`, the one pair where both arms ran to completion (bin_moment, $N_{\text{eq}} = 190$, 0.01 dpa) agrees to **nine significant figures** on $\langle n_i \rangle$ (24.563 925 879 7 Woodbury vs 24.563 925 898 6 Jacobi), seven on $\langle m_v \rangle$ and eight on swelling, with only the wall-clock differing ($1.29\times$).

2 Analysis of Results

Table 1 — the closure has converged in its own parameter

Across B2, B3 and B4 the observables move by a few percent while N_{eq} falls by a factor of ten: d_{100} spans 5.868–6.001 nm and N_{111} 3.449–3.512. That is what convergence in the closure’s refinement parameter looks like, and it is the claim Table 1 exists to support. Two qualifications are needed. First, N_{cav} and d_{cav} at B4 (3.609, 2.890) sit above B2/B3 (2.233, 2.232) by more than the loop observables move, so the cavity side is the least converged. Second, the most-resolved rung B1 never ran, so the ladder is anchored at B2 rather than at the intended top; convergence is demonstrated over a $10\times$ span in N_{eq} , not the $40\times$ planned.

Table 2 — tolerance is a null, closure order is not

$\text{rtol } 1 \times 10^{-4}$ and 1×10^{-6} agree to every digit reported. Two orders of magnitude in integration tolerance move no observable, which establishes that the tolerance is not a free parameter hiding in the results — one of the specific things the reviewer asked about. $I_{\text{bin } 10}$ likewise sits within a few percent of B4. Against that flat background, $P = 1$ (piecewise-constant closure) gives $d_{100} = 45.7$ nm against 5.9 and $d_{111} = 15.6$ against 4.6 — roughly an eightfold error — and its δ_{FP} degrades to 0.158. The existing §6.2 claim about the constant closure is therefore supported, and by a wide margin. $P = 3$ (lognormal) did not run: it reached 5.4×10^{-4} dpa of 40 with a single output step consuming 2.9 h. That is itself the result the plan anticipated, and it is reported as a failure to reach dose rather than as a number.

Table 3 — the QSS reduction is safe at fission, and the fusion column proves it

Fission QSS and fission dynamic agree to every digit: 3.533, 4.354, 4.153, 5.919, 3.253, 2.434 in both. Taken alone that flatness is uninformative — it is equally consistent with a helium treatment that does not matter and with a comparison that was never sensitive. The fusion column is what makes it a result: at the same grid, fusion QSS gives $d_{100} = 51.9$ nm and $d_{111} = 14.0$ nm, an order of magnitude away. The observables *are* sensitive to the helium environment; they are simply insensitive to how helium is time-integrated when, under fission, there is almost none to integrate ($C_{\text{He,tot}} = 1.18 \times 10^{13} \text{ m}^{-3}$ in the reference run). The fourth column, fusion + dynamic, did not reach dose, so the fusion pair is represented by QSS alone.

Table 4 — the closure reproduces the exact solution, except on cavity density

Against the fully discrete solution, the linear closure holds d_{100} to within 1.1% at every rung from 830 down to 84 equations — a $100\times$ reduction in cost for a one-percent error on that observable. N_{100} is held to 10.5% at the coarsest rung and 2% at the finest.

The exceptions are informative. N_{111} degrades sharply below C2 (−44% at C3, −51% at C4): the $\frac{1}{2}\langle 111 \rangle$ population is peaked at small sizes, and once i_{discrete} drops to 25 and then 5 that peak falls inside the binned region where the closure must represent it rather than resolve it. N_{cav} is worse still at C4 (+123%). So the coarsest rung — which is the production resolution — buys its speed at real cost in the two *density* observables, while the *sizes* remain accurate.

Convergence is not monotonic. C1 and C2 are nearly identical, C3 is worse on N_{111} and d_{100} , and C4 recovers on d_{100} while degrading further on N_{cav} . A refinement ladder that does not improve monotonically is not evidence of a converged closure, and this should be understood rather than presented as one. The most likely explanation is that i_{discrete} and the bin partition interact — the rungs hold the bin ratio r fixed but the *location* of the discrete/binned boundary moves relative to the peak of the distribution — but that has not been tested here.

$P = 1$ fails against the exact solution as decisively as it does against B4: +87% on d_{100} , −94.5% on N_{100} .

3 Effects of System Size

Figures 1–5 show the dose dependence of each observable across the ladder, ordered D, C4, C3, C2, C1, C0. d_{111} is excluded. The EUROFER97 measurements and the band spanning them are overlaid.

The curves stop at two different doses, deliberately. C2, C3 and C4 were rerun to 40 dpa so they reach the dose where the measurements sit (14.6 dpa to 32 dpa for loops). D and C1 stop at 2 dpa — the discrete arm could not be carried further at any affordable cost, and C1 was left on its original horizon.

The consequence must be read carefully: **below 2 dpa the coarse rungs can be judged against the exact solution; above it they cannot.** Where the extended curves pass through the experimental band there is no exact reference in the figure, so agreement there speaks to the model, not to the closure. The band is drawn where the data actually are so that the region with neither exact reference nor measurement cannot be mistaken for a validated one.

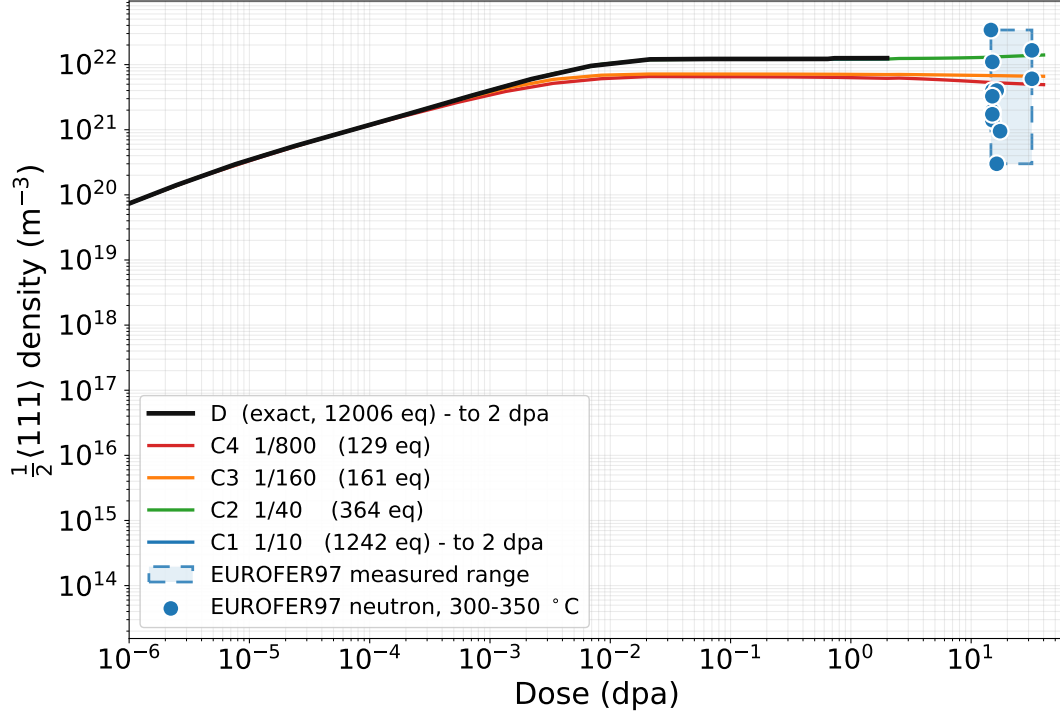


Figure 1: $\frac{1}{2}\langle 111 \rangle$ loop number density. The exact solution separates from the coarse rungs from 0.1 dpa onward; C3 and C4 lose roughly half the population.

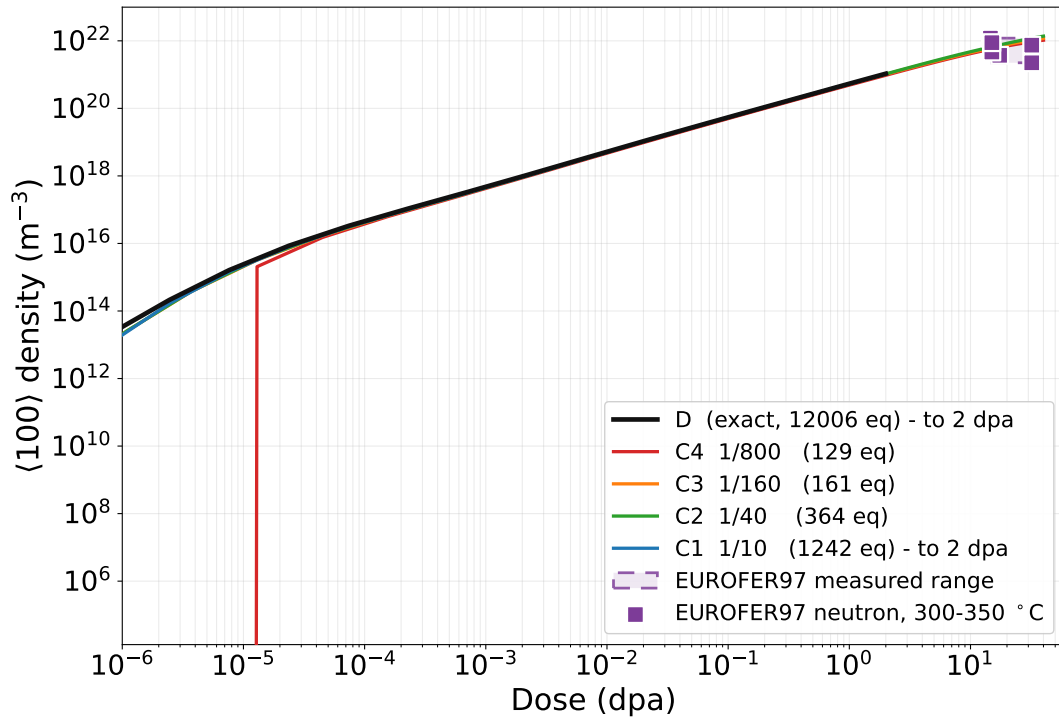


Figure 2: $\langle 100 \rangle$ loop number density. All rungs track the exact solution closely; this is the observable the closure represents best.

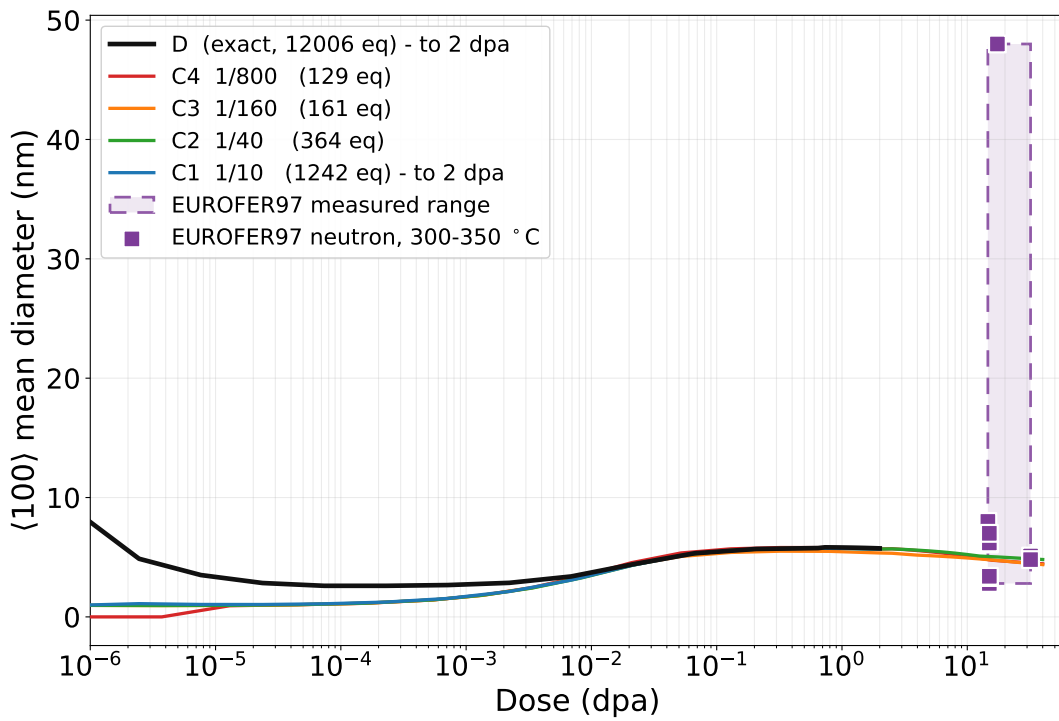


Figure 3: $\langle 100 \rangle$ mean diameter. Held to about one percent across the whole ladder.

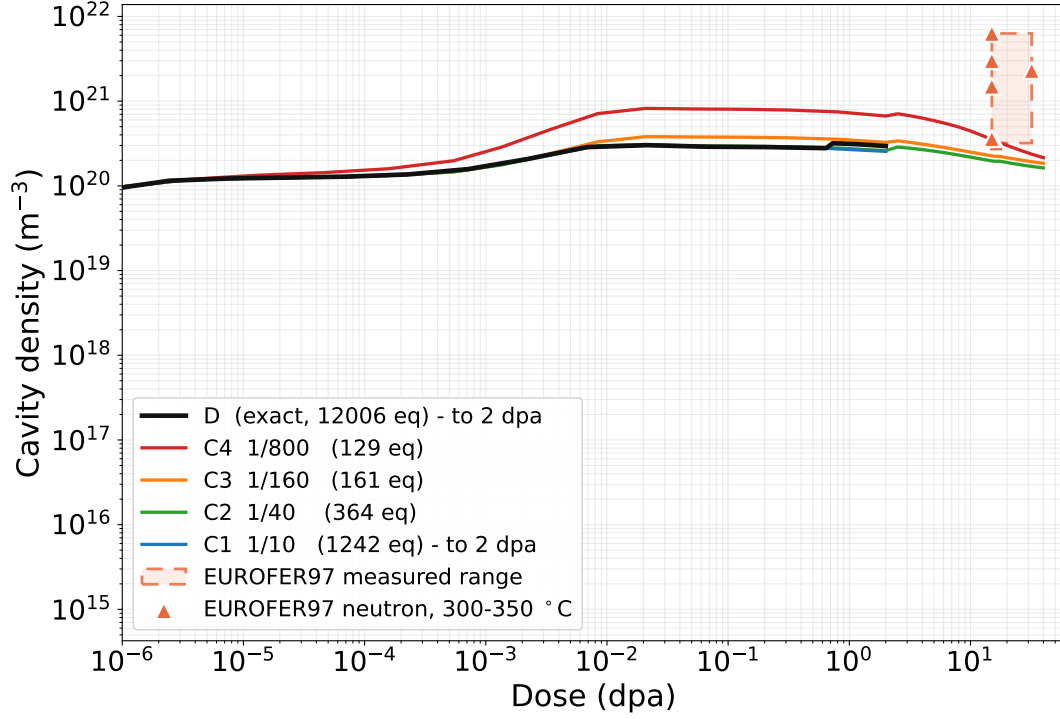


Figure 4: Cavity number density — the observable most sensitive to system size, with C4 more than a factor of two above the exact solution.

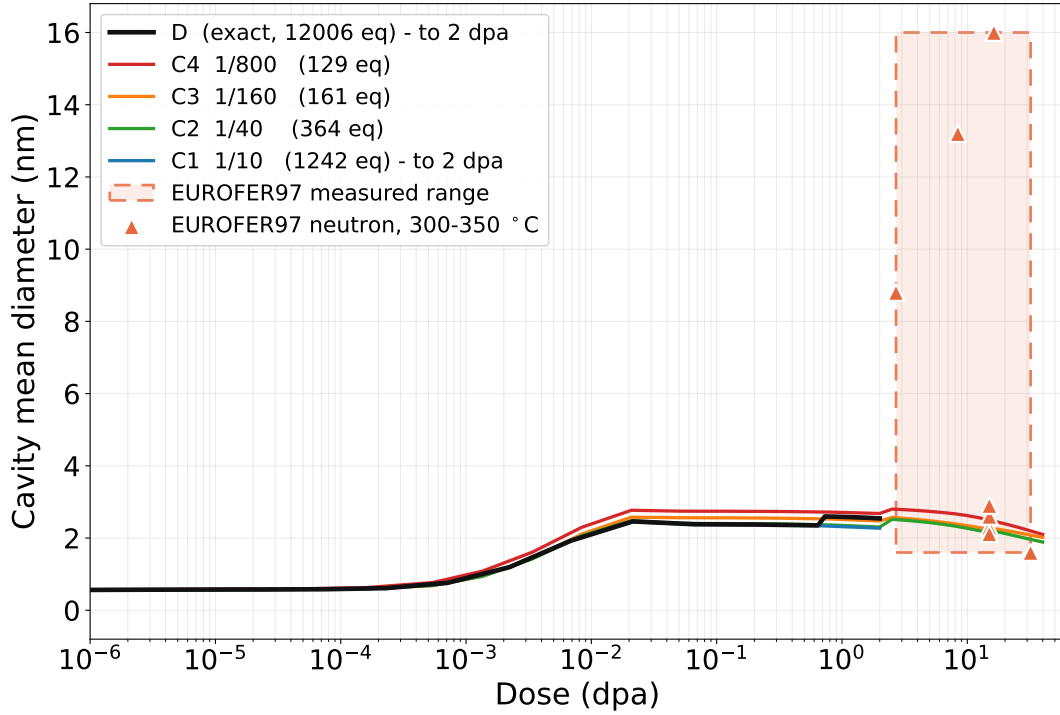


Figure 5: Cavity mean diameter. The cavity data extend down to 2.7 dpa, so this is the one panel where band and curves nearly meet.

4 Limitations

1. **Table 4 verifies a variant of the model, not the production configuration.** Production runs with `LOOP_COAL = 1`, the $\frac{1}{2}\langle 111 \rangle$ loop-loop coarsening channel. That channel was implemented only in the bin-moment solver path; the discrete path accepted the flag, recorded it in provenance, and ignored it. The discrete arm was therefore never the exact solution of the same equations, and the first version of this table — which showed a 90–114% “closure error” on d_{111} — was measuring the missing channel. The channel has since been implemented in the discrete path and the two paths verified to agree to 7×10^{-8} (`codes/Python_Testing/check_bm_vs_discrete.py`), but the production-configuration exact arm proved unaffordable: the coarsening term is $O(N_C^2)$ in the number of populated sessile sizes and, even gated, spent 2.8 h on a single output step at 0.014 dpa of 2. Table 4 is therefore run with the channel disabled in *both* arms. The closure error it measures is not demonstrably the production run’s closure error.
2. **The no-coarsening variant is unphysical.** Without the channel the $\frac{1}{2}\langle 111 \rangle$ mean size pins at the mobility cutoff and the loop density runs 20–60 $\times$ over the EUROFER97 data. This does not invalidate the verification — verification asks whether the closure reproduces the exact solution of the *same* equations, which these two arms do — but nothing in Table 4 may be compared with experiment.
3. **Four columns never reached their comparison dose:** Table 1’s B1, Table 2’s $P = 3$ and $I_{\text{bin}} 40$, and Table 3’s fusion+dynamic. Each is reported with the dose it achieved. The reporting tool refuses to print a metric for such a run, so this rule is mechanical rather than a matter of discipline.
4. $\delta_{\text{FP}} \approx 0.05\text{--}0.07$ **throughout**, against the model’s own 10^{-2} gate, and 0.158 for the two $P = 1/\text{fusion}$ columns. This is a separate, disclosed defect. A verification study measures closure error against the same equations; conservation error is not that, and the two should not be conflated.
5. **An earlier dose-ladder defect invalidated a first round of results.** The output grid held no point at the scoring dose, so a requested 15.72 dpa was silently scored at 11.51 dpa (26.8% low) while the run reported having reached it. All affected rows were recomputed. This is the same failure mode as the Figure 8 defect the study was commissioned to answer, found inside the study’s own tooling.